

Privacy-Preserving Study Space Advisor

Find the right place to study — without cameras or identity tracking.

Summer Workshop 2026 / Group [GROUP_NUMBER] / [MEMBERS]

PROBLEM & SOLUTION

PROBLEM

Students spend time searching for a space that fits their study mode. Current availability is only one part of the decision. Occupancy alone cannot describe quietness, comfort or whether discussion is appropriate. Camera monitoring adds unnecessary privacy risk.

SOLUTION

Camera-free sensing combines low-resolution thermal data, relative light, climate and non-reconstructable sound summaries. Edge processing produces anonymous room features before the backend. The dashboard presents room status, history, short-horizon forecasts and a recommendation workflow.

PROTOTYPE HARDWARE

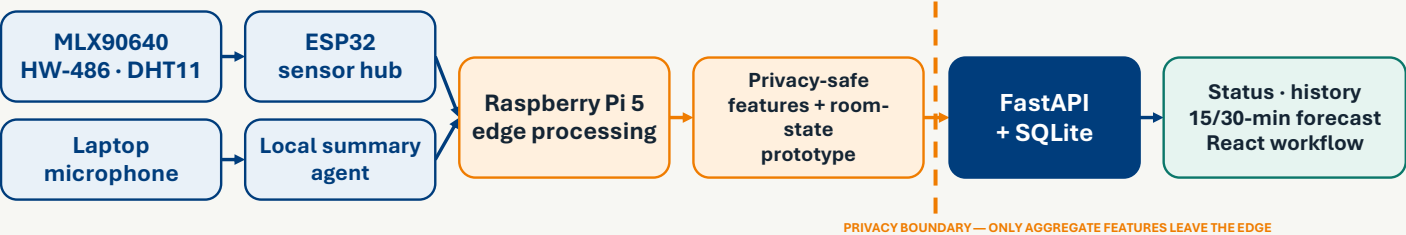
HARDWARE PANORAMA
[HARDWARE_PHOTO]

MLX90640 32×24 array
HW-486 relative light
DHT11 temperature / RH
ESP32 sensor hub
Raspberry Pi 5
Laptop microphone —
local RMS/std/peak
summaries only

PRIVACY BY DESIGN

No RGB cameras or facial recognition
No raw audio storage or upload
No identity or trajectory tracking
Thermal previews expire and are not persisted

SYSTEM ARCHITECTURE



AI & DATA INTELLIGENCE

PROTOTYPE

Room-state prototype

IMPLEMENTED

Forecast APIs

IN PROGRESS

Recommendation workflow

Anonymous summaries only — no faces, identities or speech content.

RESULTS / VALIDATION

4

real scenario sessions

36

five-second windows

176

MLX90640 thermal frames

Gate A automated end to end
Real sensor summaries captured
Four sessions privacy-validated
Independent evaluation in progress

LIVE DEMONSTRATION • UI PROTOTYPE

Quiet Commons

32 × 24 s

Thermal view · mock data

Low occupancy0.7

Brightness0.3

Temperature and humidity0.4

Current state matches quiet study; Current occupancy is low.

#2 Discussion HubMedium occupancy, 30m MediumGood

Recommendation UI · mock data

